Fiscal Year:	FY 2012	**Task Last Updated:**	FY 09/06/2012
PI Name:	Brandizzi, Federica Ph.D.		
Project Title:	Role of endomembranes in plant responses to space environment stresses		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	48824-1046	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	05/27/2009	**End Date:**	08/26/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/26/2012 per NSSC info (Ed., 6/02/2012)		
Key Personnel Changes/Previous PI:	PI: Federica Brandizzi		
COI Name (Institution):			
Grant/Contract No.:	NNX09AK90G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Successful plant growth in closed-loop life support conditions is a difficult challenge for the realization of long-term habitation of spacecraft and other extraterrestrial environments. In such environments, plants can undergo stress induced by a number of factors including microgravity, vibration, limited exchange of gases, and suboptimal growth conditions (temperature, light, nutrients). This results in limited plant growth, development and yield. Fortunately, through genetic engineering plants can be coaxed by humans to adapt to hostile environments. To take full advantage of this ability, we first have to understand how plants respond to stresses. Plants have evolved elaborate surveillance and response systems consisting of a large repertoire of sensors and transducers to cope with stress. Fast stress management responses include activation of membrane-tethered transcription factors (MTTFs), which are dormant transcription factors when first synthesized. Following stress sensing, MTTFs are quickly released from the membrane anchor and re-targeted to the nucleus to invoke downstream responses. Although we have a wealth of knowledge of MTTFs in non-plant systems, it is still unclear how these factors function in plants. What we know, however, is that a handful of MTTFs of the AtbZIP family are anchored to the endoplasmic reticulum membrane and are activated in conditions of stress. Yet we do not have a complete picture of the type of stress that these AtbZIP-MTTFs respond to, nor of the mechanisms for their activation and stress transduction. Here, we propose to use cell biology, biochemistry, and genetics in the model plant Arabidopsis thaliana to address these important questions. The results gathered in our research will contribute to NASA's strategic plans for the realization of the long-term habitation of space and planetary surfaces. A better understanding of stress sensing, integration, and response mechanisms in plants will make possible the design and growth of plants resistant to space stresses, and of plants that function as stress biosensors. Development of our research will also contribute to the characterization of basic signaling pathways including cross-talk between the endomembrane system and the genome.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding how plants perceive and respond to stress is important to optimize them to survive in a variety of habitats and for the colonization of new ones. We anticipate that our research results will contribute to NASA’s strategic plans for the realization of the long-term habitation of space and planetary surfaces but also colonization of hostile and underused environments on Earth. A better understanding of stress sensing, integration, and response mechanisms in plants will make possible the design and growth of plants resistant to the stress of space travel and habitation, as well as hostile habitats on Earth, and of plants that function as stress biosensors. Development of our research will also contribute to the characterization of basic signaling pathways including cross-talk between the endomembrane system and the genome.
Task Progress:	Our project focuses on the dissection of the mechanisms that plants use to resist to stress with particular attention to signal transduction mediated by membrane-anchored sensors using techniques of cell biology, biochemistry, and genetics in the model plant Arabidopsis thaliana. Similar to animals, plants suffer from stress linked to a variety of challenges. For example, certain adverse environmental conditions, such changes in temperature, reduced nutrients and disease, as well as physiological situations, such as growth, can generate an imbalance between demand and capacity of protein synthesis in the endoplasmic reticulum (ER), a condition that is also known as ER stress (Ron and Walter, 2007; Liu and Howell, 2010). The ER is the organelle that initiates the production of all secretory proteins. Secretory proteins constitute one third of the cellular proteome and allow cells to communicate with the external environment. The ER can sense stress due to accumulation of unfolded proteins in its lumen, and restores homeostasis by invoking a largely conserved cytoprotective signaling pathway, called the unfolded protein response (UPR). Compromised UPR leads to serious conditions and even death in animals and plants (Iwawaki et al., 2009; Deng et al., 2011; Chen and Brandizzi, 2012). In animals and plants, several sensors and transducers initiate the UPR; however, IRE1, an ER-associated protein kinase and ribonuclease, functions as a major ER stress sensor and plays a key role in the UPR signaling by controlling UPR gene expression (Tirasophon et al., 2000; Urano et al., 2000; Chen and Brandizzi, 2012). During our current award, we have advanced fundamental knowledge of the plant UPR. In particular, in addition to showing that AtIRE1 is a key regulator of UPR in Arabidopsis, we have also demonstrated that AtIRE1 controls root elongation in normal conditions of growth, thus revealing a physiological role to AtIRE1 in development (Chen and Brandizzi, 2012). We have been also able to demonstrate that AtIRE1 controls the expression of another UPR regulator, AGB1, which is a subunit of the ubiquitous heterotrimeric GTP-binding protein family (Jones and Assmann, 2004). In addition, through genetic studies, we have established that AGB1 and AtIRE1 independently control UPR (Chen and Brandizzi, 2012). These results are the first example of a genetic interaction between UPR regulators in plants (Chen and Brandizzi, 2012). Through a collaborative effort with colleagues in the US and South America, the UPR mutants generated in our lab have been used to provide evidence for an involvement of the two AtIRE1 isoforms in the activation of the transcription factor bZIP60 in response to chemically-induced stress, as well as to pathogens (Moreno et al., 2012). In a separate effort, we have also gathered evidence that the UPR is activated in conditions of changed gravity. Pursuing this aspect of the research is our current priority in a new NASA award to our lab (Grant/Cooperative Agreement Number: NNX12AN71G). Finally, in collaboration with Dr. S. Howell (Iowa State Univ.), we have also characterized the determinants necessary for the trafficking and activation of a membrane tethered transcription factor in response to stress (Srivastava et al., 2012). Together our efforts are providing fundamental insights in the understanding of the regulatory network for growth as well as for stress sensing and response in plants. Our award has led to the publication of three peer-reviewed manuscripts in top journals (Chen and Brandizzi, 2012; Moreno et al., 2012; Srivastava et al., 2012) as well as to the training of a PhD student, Ms. Yani Chen (graduation Spring 2013). Literature (* indicates publications from this award): *Chen, Y., and Brandizzi, F. (2012). AtIRE1 and AGB1 independently control two essential unfolded protein response pathways in plants. Plant J. 69, 266-277. Deng, Y., Humbert, S., Liu, J.X., Srivastava, R., Rothstein, S.J., and Howell, S.H. (2011). Heat induces the splicing by

IRE1 of a mRNA encoding a transcription factor involved in the unfolded protein response in Arabidopsis. Proc Natl Acad Sci U S A 108, 7247-7252.

Iwawaki, T., Akai, R., Yamanaka, S., and Kohno, K. (2009). Function of IRE1 alpha in the placenta is essential for placental development and embryonic viability. Proc Natl Acad Sci U S A 106, 16657-16662.

Jones, A.M., and Assmann, S.M. (2004). Plants: the latest model system for G-protein research. EMBO Rep 5, 572-578.

Liu, J.X., and Howell, S.H. (2010). Endoplasmic reticulum protein quality control and its relationship to environmental stress responses in plants. Plant Cell 22, 2930-2942.

*Moreno, A.A., Mukhtar, M.S., Blanco, F., Boatwright, J.L., Moreno, I., Jordan, M.R., Chen, Y., Brandizzi, F., Dong, X., Orellana, A., and Pajerowska-Mukhtar, K.M. (2012). IRE1/bZIP60-Mediated Unfolded Protein Response Plays Distinct Roles in Plant Immunity and Abiotic Stress Responses. PLoS One 7, e31944.

Ron, D., and Walter, P. (2007). Signal integration in the endoplasmic reticulum unfolded protein response. Nat Rev Mol Cell Biol 8, 519-529.

*Srivastava, M., Chen, Y., Deng, Y., Brandizzi, F., and Howell, S.H. (2012). Critical elements in bZIP28 mediate its organelle-to-organelle movement under ER stress conditions. Plant J. 70, 1033-1042.

Tirasophon, W., Lee, K., Callaghan, B., Welihinda, A., and Kaufman, R.J. (2000). The endoribonuclease activity of mammalian IRE1 autoregulates its mRNA and is required for the unfolded protein response. Genes Dev 14, 2725-2736.

Urano, F., Bertolotti, A., and Ron, D. (2000). IRE1 and efferent signaling from the endoplasmic reticulum. J Cell Sci 113 Pt 21, 3697-3702.

Bibliography Type:	Description: (Last Updated: 12/08/2025)
Articles in Peer-reviewed Journals	Chen Y, Brandizzi F. " AtIRE1A/AtIRE1B and AGB1 independently control two essential unfolded protein response pathways in Arabidopsis." Plant J. 2012 Jan;69(2):266-77. Epub 2011 Oct 21. http://dx.doi.org/10.1111/j.1365-313X.2011.04788.x ; PMID:21914012 , Jan-2012
Articles in Peer-reviewed Journals	Moreno AA, Mukhtar MS, Blanco F, Boatwright JL, Moreno I, Jordan MR, Chen Y, Brandizzi F, Dong X, Orellana A, Pajerowska-Mukhtar KM. "IRE1/bZIP60-mediated unfolded protein response plays distinct roles in plant immunity and abiotic stress responses." PLoS One. 2012;7(2):e31944. Epub 2012 Feb 16. PubMed PMID: 22359644 ; http://dx.doi.org/10.1371/journal.pone.0031944 , Feb-2012
Articles in Peer-reviewed Journals	Srivastava R, Chen Y, Deng Y, Brandizzi F, Howell SH. "Elements proximal to and within the transmembrane domain mediate the organelle-to-organelle movement of bZIP28 under ER stress conditions." Plant J. 2012 Jun;70(6):1033-42. http://dx.doi.org/10.1111/j.1365-313X.2012.04943.x ; PMID:22335396 , Jun-2012